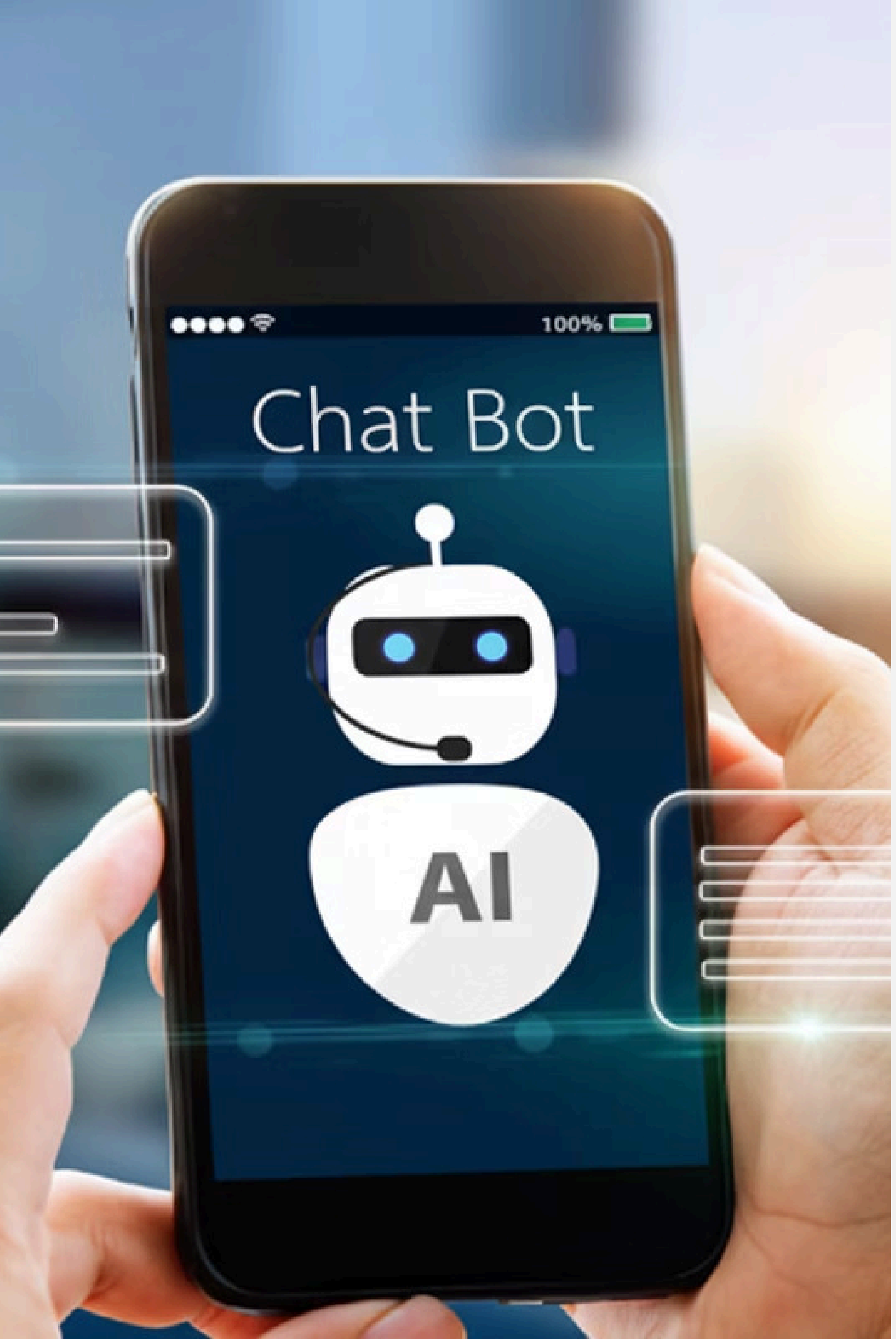

IIPC Assistant

- **Title:** IIPC Assistant: An AI-Powered Web Archiving Platform

- **Name:** Youssef Wael Nabil
- **Student ID:** 21100877
- **Department:** Artificial Intelligence, Faculty of Computer Science & Engineering
- **University:** Al Alamein International University

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Agenda

- The Problem
- Aim
- Why RAG? (Architectural Decision)
- System Architecture
- Development Journey: Milestones 1 & 2
- Optimization
- Conclusion & Future Work
- The Final Application (Demo)

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Problem

Valuable IIPC resources (conference posters, presentations) are hard to access and search.

Traditional search lacks **semantic understanding**.

Extracted text from PDFs/Video is **unstructured and noisy** (OCR errors, fragmented slides).

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Example

raw text extracted from conference presentation slides:

```
Mark Phillips, UNT Libraries \nAbbie Grotke, Library of Congress \n@vphill |
mark.phillips@unt.edu\n@agrotke | abgr@loc.gov\nThe End Of Term Archive:
Collaboratively Preserving the United States \nGovernment Web\nNovember 14, 2018\n
\nit all began a long, long, time ago, in a far away place
\nhttps://flic.kr/p/4N2jHU\nhttps://flic.kr/p/4JNkLE\n(to the United States, not
here)\n2\n\noriginal end of term web archive partners\nfor 2008/2012/2016 - all
IIPC & NDIIPP/NDSA partners\nNovember 14, 2018\nIIPC - WAC 2018\n3\n\nextant gov
web archiving efforts\nCapture, Preservation, & Access\n•\nLOC: .gov, election,
other\n•\nGPO: agency sites, often ephemeral \n•\nNARA: congressional web harvest
every \n2 years\n•\nIA: global & curated crawls\n•\nAgency-level: NIH/NLM, DOE,
DOL, HHS, \nCMS, others, using Archive-It or other \ntools\n•\nUNT & Others:
Topical .gov collecting\n•\nInternal agency guidelines\nCommunity Efforts\n•
\nFederal Web Archiving Group\n•\nmost of those at left plus other feds\n•
\nResearch Initiatives\n•\nacademic\n•\nNGO or watchdog\n•\nCitizen Driven\n•
\ngrassroots efforts\n•\nEnd of Term\n•\nfocused but large-scale
multi-\ninstitutional project\nNovember 14, 2018\nIIPC - WAC 2018\n4\n\nwork
collaboratively to preserve public U.S. Government websites\ndocument federal
agencies’ presence on the web during the end of Presidential \nterms\nenhance the
existing research collections of the partner institutions\nraise awareness about
the need for preservation\nengage with researchers and subject experts\ngoals of
the end of term project\nNovember 14, 2018\nIIPC - WAC 2018\n5\n\neot
collaborative distribution of work\n•\nIA: crawling, preservation, access, full-
text \nsearch, metadata, research tool & support\n•\nLC: crawling, preservation,
```

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Aim & Objectives

- **Primary Aim:** To build an AI-powered platform for natural-language querying of IIPC materials.

Knowledge Base

Searchable repository of 587+ documents

Semantic Search

Vector embeddings for relevance-ranked retrieval

AI ChatBot

Context-aware assistant with verifiable sources

Modern UI

Responsive web interface for seamless access

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Available data

- **Volume:** 587 IIPC documents harvested via OAI-PMH from UNT Digital Library.
- **Types:** Conference posters, presentation slides, and video transcripts.
- **Challenge:** The extracted text was noisy and poorly structured.
- **Specific Problems:** OCR errors and formatting artifacts from PDFs.

IIPC_materials extracted:

- **title** – text field for the material's title
- **creator** – text field for the author or contributor
- **date** – date of the material
- **description** – detailed text description
- **subject** – topic or keywords
- **item_type** – type of item (e.g., paper, presentation, video)
- **ark_url** – link to the material
- **source_url** – link to the item
- **content** – extracted text or content

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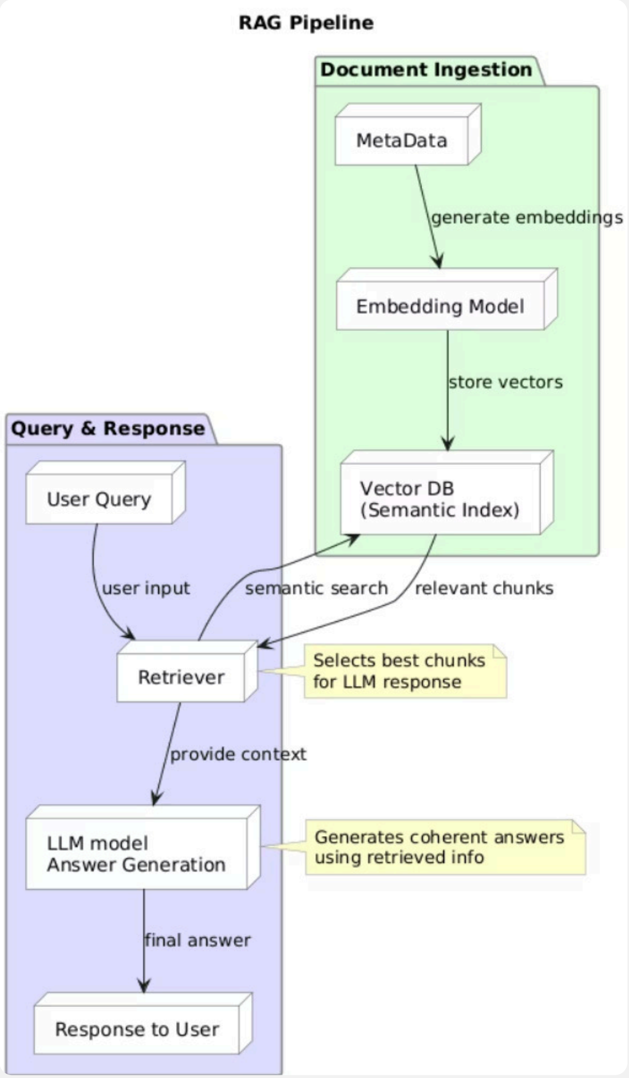
Why RAG?

Challenge: Dataset is unstructured and lacks QA pairs for fine-tuning.

- **Solution: Retrieval-Augmented Generation (RAG)**
- **Advantages:**
  - **Scalable:** Add new content without retraining.
  - **Transparent:** Every answer is based on grounded sources.



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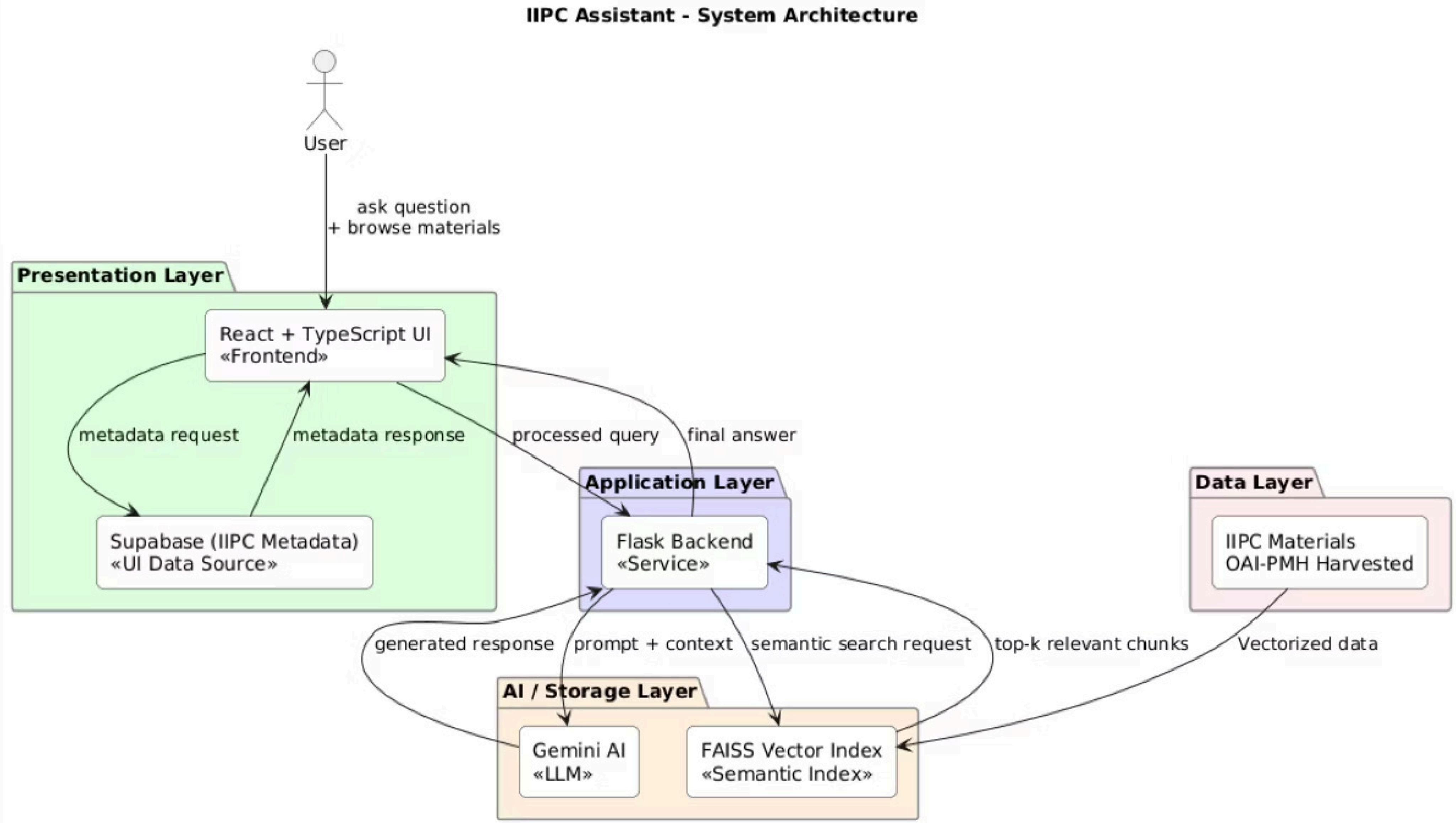
System Architecture



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Milestone 1:

Accomplishment 1: Data Harvesting

- Automated collection of **587 documents** via OAI-PMH.
- Preserved full metadata (title, creator, date, ARK URL).

Accomplishment 2: Tech Stack Selection

- Embeddings: `BAAI/bge-base-en-v1.5`
- LLM: `tiiuae/falcon-7b-instruct` (Initial)

Challenge Identified: Raw text quality was poor, limiting answer accuracy.

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Results after Milestone 1

```python
query = "what is the difference between crawling and harvesting ?"
query_embedding = embed_query(query)
top_ids = search_index(query_embedding, index)
top_chunks = chunked_df.iloc[top_ids]["text"].tolist()

prompt = build_prompt(query, top_chunks)
response = llm(prompt, max_new_tokens=100, do_sample=True)[0]["generated_text"]
print(response)
```

```
Batches: 100% 1/1 [00:00<00:00, 49.91it/s]
Setting `pad_token_id` to `eos_token_id`:11 for open-end generation.
Use only the context below to answer the user's question.
    If the answer is not in the context, say "I don't know".

    Context:
    - : selection side Letting students constitute ready harvesting ncorpus Groups students Corpus manageable size: URLs Choosing subject theme significant specific presence web Allowing selection different website
s ( editorial point nview) Related .. recent event local topic Defining URL Relevant descriptors (specific chosen group) Harvesting parameters (depth frequency) Filling spreadsheet informationnPractical experience
: harvesting side Review basics (HTML, URL, HTTP) sites narchitectures Presentation different archiving methods server side transactional client side work advantagesdrawbacks focus crawlers (exercizing HTTrack) sof
tware companies, profits, online services formats (exercizing wget)nPractical experience : access side harder training? hardly explain students install Wayback Machine! Viewing selected websites Wayback

- basically now. separate cluster. scrape metadata endpoints, URLs, scrape things institutional repositories other content sources crawl those maybe every three months maybe twice something those crawls often month
s hundreds millions other stuff course global crawls internet archive already which collect couple billion every year. wanted those crawling though didn getting intentionally tools through processes intentional acq
uisition projects. lastly, journals other websites crawl, which think know. wanted least publishers authors scholars themselves ability deposit things couldn because maybe behind paywall, there version preprint aut
hor version something that. ability individuals institutions collection. categorized top known approach where harvesting metadata, registries, etc., etc.,

- 2012nGoals workshop about yours?n08 2012n10nMain definitions: processes Crawl nprocess browsing using dedicated software nautomatically recursively following links allowing ntheir captured differentiated crawls s
elective crawls Crawl instance single process launched crawling robot delivers based several seeds crawl nsettings Target seeds along harvesting crawl settings (frequency, scope, robots policy, budget, etc.) ndesig
ned crawl meaningful contentn08 2012n11nMain definitions: outcomes Archive entire institution webharvested material, ncomprising collections Collection cohesive grouping content which makes sense ngrouping accordin
g collecting institution policy Capture instance given website being harvested through ngiven processn08 2012n12nCollection development statistics Measure volume general development nArchive number documents size.
nmeasure quantitative output collecting process

- the harvest material within selected spann27nDomain nDomain nDomain nDomain nB1 nDomain nB2 nD1 nDomain nSTUDYING NATION DOMAIN TIMEnNiels Brügger, Ditte Laursen Janne Nielsenn27 APRIL 2015nCORPUS CREATIONnTest a
lgorithmnTested first broad crawl January 2006 (1TB, websites 10MB)nThis harvest consists 127 jobsnEach consist several domains produce 18GB crawl enhanced IDsn28nSTUDYING NATION DOMAIN TIMEnNiels Brügger, Ditte La
ursen Janne Nielsenn27 APRIL 2015nCORPUS CREATIONnTest algorithmnUsing BigInsights perform algorithm large spreadsheet algorithm locates objects nincluded harvest (harvests)nThere might duplicates these cases, nalg
orithm identifies selects objects closest harvestn29nSTUDYING NATION DOMAIN TIMEnNiels Brügger, Ditte Laursen Janne Nielsenn27 APRIL 2015nNEXT STEPSnFrom implementation crawl material crawl refer analyze? Should fi
les opened?

- platform. That about terabytes data, which about sixth crawl every year. really continuation doing blogs about years. committed track blogs transformed personal writing, personal diaries. professional personal co
mmitment among colleagues. already harvested blogs years ago, tens, thousands, really scale. possible, started whoa. really challenge, although domain crawls. really, strategy relied cooperation, compromise, improv
ement crawl processes. between August year, worked closely teams. today, Maxime Dequey, Patrice Damzin, Franck Chénaud. meeting first which starting point right harvest start always easier discover small tests star
ted blogs hours never really ended first estimation complete harvest about beyond years. quite solutions. started defining would acceptable

    Question: what is the difference between crawling and harvesting ?
    Answer:
Crawling is a process of collecting information from a website by following links from one page to another. It is a linear process where a spider starts at the top of a website and scrapes the information from each
page. The goal of crawling is to extract information from a website by following links from one page to another.

Harvesting is a process of collecting information from multiple websites at once. It is an automated process where a spider follows links from one page to another on
```

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Milestone 2

Accomplishment 1: AI-Powered Text Enhancement

- Used **Google Gemini 2.5 Flash-Lite** to restructure all documents.
- Transformed messy text into coherent narratives.
- **Result:** A high-quality, semantically rich dataset.

Accomplishment 2: Functional Prototype

- Built a Streamlit app for testing.
- Then build web interface
- Upgraded LLM to **Google Gemini** for generation.
- **New Challenge:** Retrieval needed optimization for broader context.

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Text Enhancement

```
---
# **The End of Term Archive: Collaboratively Preserving the United States Government Web**
*Presented by:**
**Mark Phillips**
UNT Libraries
@vphill | mark.phillips@unt.edu
**Abbie Grotke**
Library of Congress
@agrotke | abgr@loc.gov
*Date:** November 14, 2018
*Event:** IIPC - WAC 2018
---
## **1. Introduction and Context**
**Inspiration:** "It all began a long, long, time ago, in a far away place..." (referring to the United States)
[https://flic.kr/p/4N2jHU](https://flic.kr/p/4N2jHU)
[https://flic.kr/p/4JNkLE](https://flic.kr/p/4JNkLE)
**Original End of Term Web Archive Partners:**
Encompassed IIPC & NDIIPP/NDSA partners for the 2008, 2012, and 2016 cycles.
---
## **2. Existing Government Web Archiving Efforts**
*A. Capture, Preservation, & Access Initiatives:**
**Library of Congress (LOC):** Focuses on `.gov` domains, election-related content, and other areas.
**Government Publishing Office (GPO):** Archives agency sites, particularly those with ephemeral content.
**National Archives and Records Administration (NARA):** Conducts biennial web harvests of congressional websites.
**Internet Archive (IA):** Manages global and curated web crawls.
```

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Results after Milestone 2

Youssef Eldakar

Feedback form within the IIPC...

If this is "just a feasibility study", I'm wondering what you have in mind for the final product :slightly_smiling_face: In short, the Assistant has surpassed my expectations! I'd like to learn more about how it was built and how it's actually working.

I've tried a bunch of different searches and it's very good with some content (e.g. questions about specific tools). So far, I've only have 2 inaccurate / not fully accurate answers. Helpfully, it says "I don't know" if there's no answer rather than making it up which is what other bots do.

8:35 PM

And...

Here's one interesting example which I think shows why the model trained on the WA data should be more precise:

8:38 PM

AI Overview

Web crawling and web scraping are related but distinct processes for gathering data from the internet. Web crawling, also known as web spidering, is the process of systematically discovering and indexing web pages by following hyperlinks. Web scraping, on the other hand, is the focused extraction of specific data from those web pages. In essence, crawling finds the pages, and scraping extracts the data.

AI response

What is the difference between crawling and harvesting?

13:23

IIPC Assistant

The provided documents do not explicitly define the difference between "crawling" and "harvesting." However, they discuss these terms in the context of web archiving.

Crawling is described as the process of fetching URIs and discovering additional URIs, an iterative process that continues until a "done" condition is met. It involves agents performing data collection.

Harvesting is discussed in relation to meeting the "harvesting needs of the preservation community" and as "web harvesting toolkit." It is also referred to as "declarative harvesting" where the user declares their harvesting goals.

Our chatBot response

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Optimized Retrieval Method and Prompt

Old Method	New Method
• Retrieve top-k chunks by score. • Risk: Gets details but misses the big picture."	1. Retrieve top 30 chunks. 2. Group by Document ID. 3. Rank documents by their best chunk. 4. Select top 3 documents. 5. Take best chunks from them.

Optimized Prompt: Summarize IIPC conference content using correct web archiving terms and always include where it came from (title, author, date, ARK URL)."

Conclusion: Ensures answers are comprehensive and document-aware.

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Results after Optimization

what is the difference between crawling and harvesting ?

10:04 PM

IIPC Assistant

Based on the available IIPC materials, the terms "crawling" and "harvesting" are used in the context of web archiving, and while closely related, they refer to distinct aspects of the process.

Crawling refers to the process of agents performing data collection on the web. These agents, known as crawlers, start with initial URLs (seeds) and navigate the web to identify and retrieve content. The frontier is the managed list of URLs to be visited during this process. Alex Dempsey's presentation "Better Together: Building a Scalable Multi-Crawler Web Harvesting Toolkit" (IIPC WAC 2025) outlines "crawlers" as the "agents performing the data collection." Jefferson Bailey's presentation "Crawlward Ho!" (IIPC GA 2016) also discusses "current crawling paradigms" such as archival link-based crawling, browser-based crawling, transactional crawling, and API-based crawling. Marius André Elsfjordstrand's presentation "Lessons Learned Building a Crawler From Scratch: The Development and Implementation of Veidemann" (IIPC WAC 2025) details the development of "Veidemann, a browser-based web crawler."

Harvesting, on the other hand, is the broader activity of collecting web content for preservation. Alex Dempsey's presentation "Better Together: Building a Scalable Multi-Crawler Web Harvesting Toolkit" (IIPC WAC 2025) describes it as meeting "the harvesting needs of the preservation community" and emphasizes a "scalable, multi-crawler, and multi-tool strategy for web harvesting." Marius André Elsfjordstrand's presentation "Lessons Learned Building a Crawler From Scratch: The Development and Implementation of Veidemann" (IIPC WAC 2025) discusses a move towards "declarative harvesting, where the user declares their harvesting goals, and the system orchestrates the process to achieve them." This implies that harvesting is the overarching goal, and crawling is a method or component used to achieve that goal.

In essence, crawling is the mechanism by which web content is discovered and retrieved, while harvesting encompasses the entire process of collecting this content for digital preservation purposes, often involving multiple crawling strategies and tools.

10:04 PM

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INTERNATIONAL UNIVERSITY
جامعة العلمين الدولية

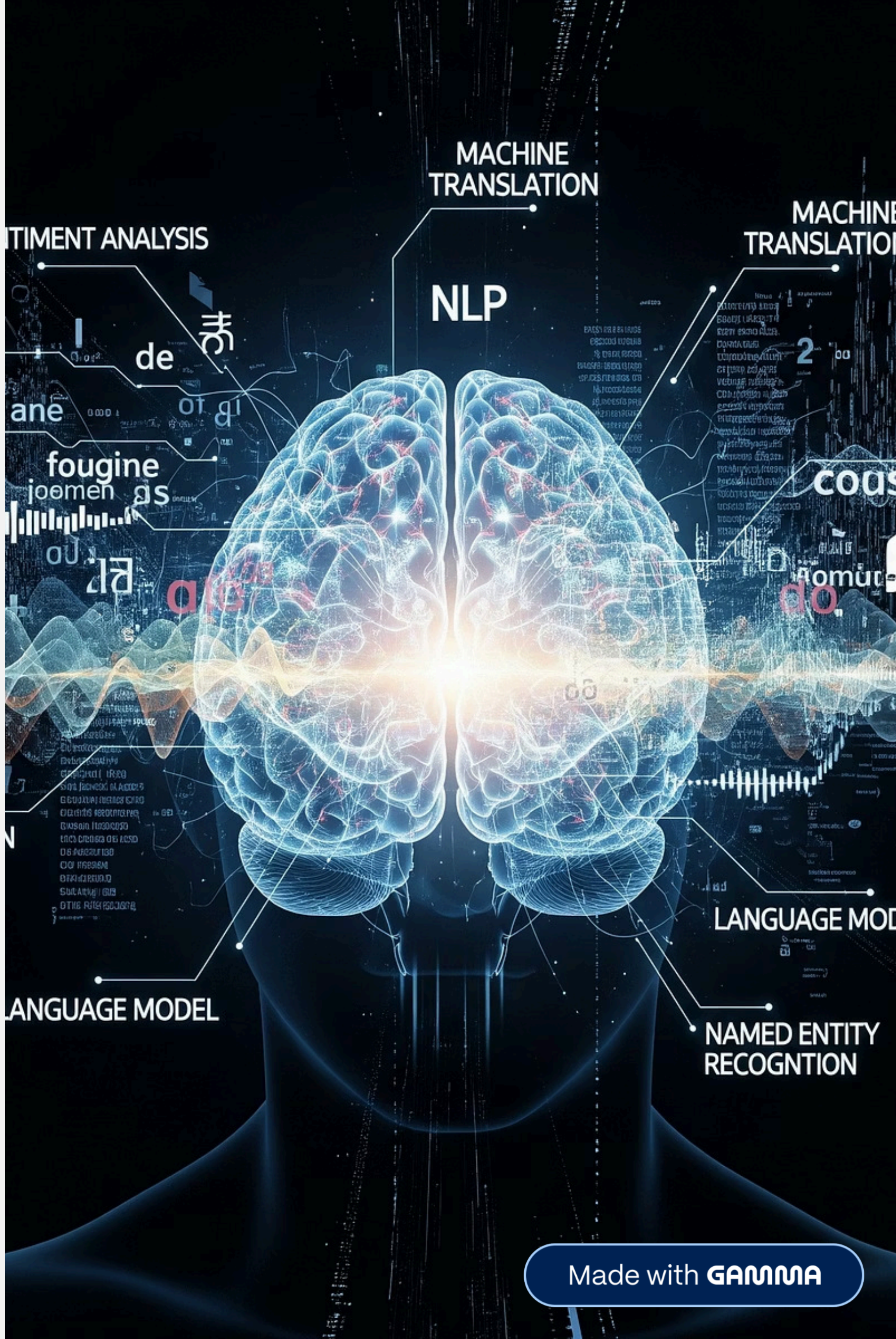
Conclusion & Future Work

Conclusion: Successfully delivered an AI platform that significantly enhances access to IIPC's specialized knowledge.

Key Innovations: AI Text Enhancement, Document-Grouped Retrieval, Domain-Specific RAG.

Future Work:

- Expand document collection.
- Add user accounts and query history.



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Demo

Google Docs

Demo.mp4

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